classmate
Date
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ASSIGNMENT

Real Time scheduling Algorithms

Task	Capacity	Period	Deadline	Absolute deadline
t_1	3	20	7	7
t_2	2	5	4	4, 9, 14, 19
t_3	2	10	8	8, 18

Hyper period = LCM(20,5,10) = 20 Utilization = 3/20 + 2/5 + 2/10 = 0.75

Q1 Rate monotonic Scheduling (RMS)

a) What is the main idea behind RMS? How do they decide which task gets the highest priority?

b) Give the priority order for the task above

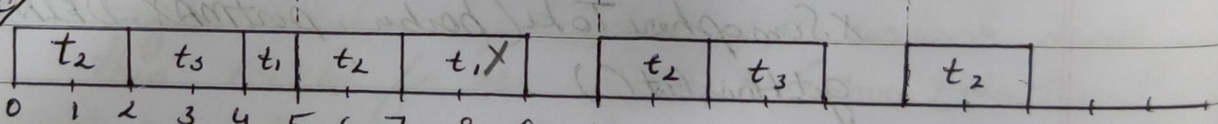
c) Draw the schedule for one hyper period (all task start at time 0, premption allowed) Does every task meet deadline? If not explain why it fail.

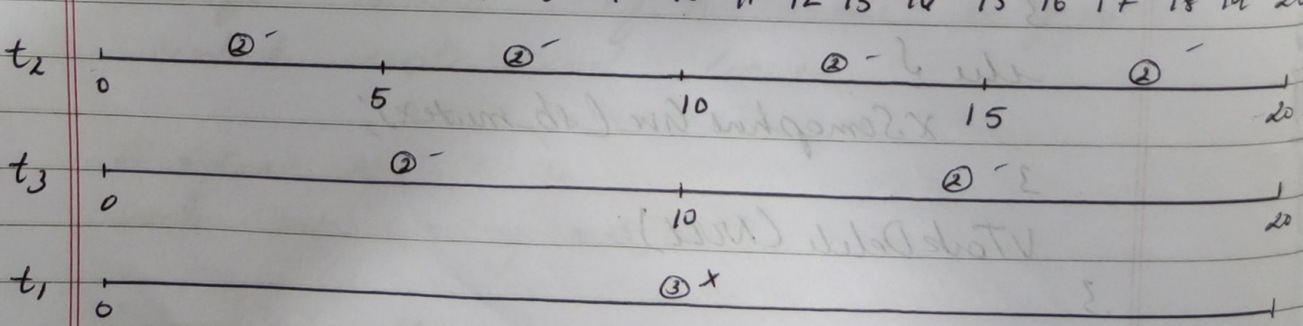
Ans

a) RMS is a fixed priority algorithm. The task with ~~the highest~~ shortest period (most frequent) get the highest priority, and priority never change. A running task is prempted whenever higher priority task is ready.

b) Priority order $t_2 (T=5) > t_3 (T=10) > t_1 (T=20)$

c)





t1 miss the deadline: It finishes at 9 but deadline is 7. RMS looks only at the period, so it treats t1 as the least urgent task. But t1 has tightest deadline (7).

~~Since t2 and t3 take CPU for~~

The utilization test (0.75 < 0.78) the bound for 3 task would wrongly suggest that the test is schedulable

Q2 Deadline Monotonic scheduling (DMS)

a) What is the main idea behind DMS. When is it a better choice than RMS

b) Give the priority order for the task above. Is it the same as RMS? Explain why or why not.

c) Draw the schedule for one hyperperiod. Does every task meet its deadline? If not why it fails.

Ans

a) DMS is also a fixed priority scheduling but the task with the shortest deadline gets the highest priority. It is better than RMS when deadline are shorter than period ($D < T$). It see how urgent the task really is. When $D = T$ DMS & RMS give same order

b) Priority order

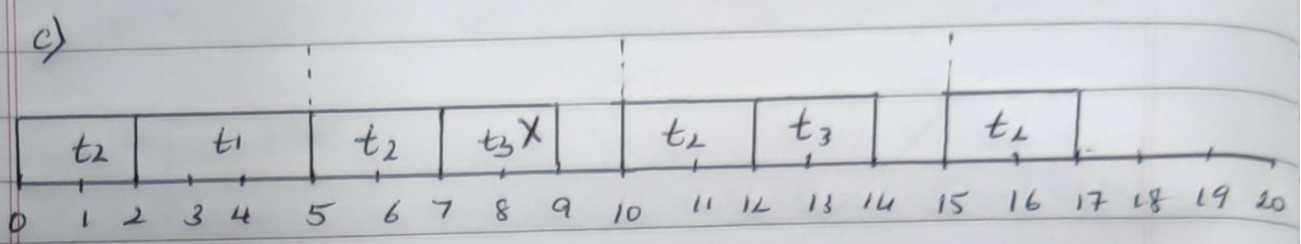
$t_2 (D=4) > t_1 (D=7) > t_3 (D=8)$

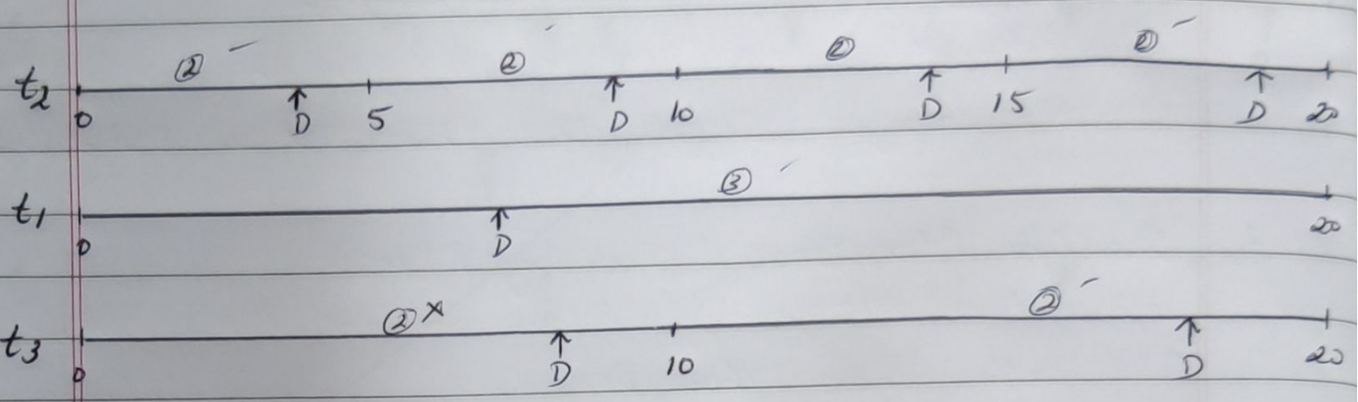
This is not the same as RMS.

RMS puts t3 above t1 because t3 has the shorter period.

DMS put t1 above t3 because t1 has Shorter deadline.

c)





t_3 misses the deadline by 1 unit. Fixing t_1 helped but t_3 is now lowest priority. It waits for t_2 then t_1 then t_2 again only starts at 7, too late to finish by 8. DMS is better than RMS with fixed priority still loses 1 task.

Q3 Earliest deadline first scheduling (EDF)

a) What is the main idea behind EDF? How is it different from RMS and DMS, where priorities stay fixed

b) Draw the schedule for one hyperperiod. At each decision point say which task is picked & why

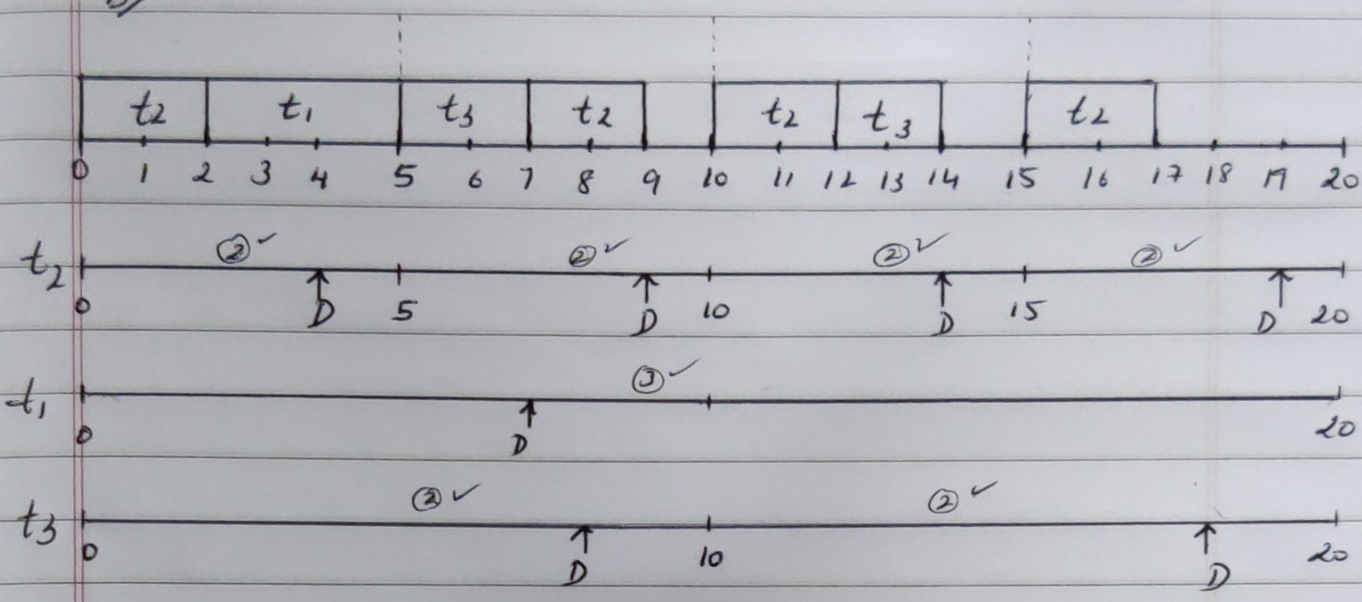
c) Does any task meet deadline? Why does EDF succeed when fixed priority method struggle

Ans

a) EDF is a dynamic priority algorithm. At every decision point (a job releases or finishes), the CPU goes to the ready job with the nearest absolute deadline. Priorities are not tied to tasks so they

change from job to job. In RMS & DMS the order is fixed in advance

b)



Time line

t_2 (0-2) → t_1 (2-5) → t_3 (5-7) → t_4 (7-9) → idle (9-10)
→ t_2 (10-12) → t_3 (12-14) → idle ((14-15) → t_2 (15-17) → idle (17-20)

c) Every task meets the deadline
EDF succeed because at time 5 it saw t_3's deadline 8 was near than new t_2 job deadline (9) so it ran t_3 first.
A fixed priority scheme cannot make that choice because t_2 always outranks t_3
EDF spends CPU on most urgent task.

Comparison

Algorithm	Priority based on	Fixed/Dynamic	Result for this set
RMS	Period	Fixed	t_1 miss
DMS	Relative Deadline	Fixed	t_3 miss
EDF	Absolute Deadline	Dynamic	All deadlines met